

SCRAPFLOW AI

Complete Founder Blueprint v1.0

Startup-Ready Product, Business & Technical Specification

1. Executive Summary

Product Name

ScrapFlow AI

Product Type

Mobile-First SaaS Platform for Scrap Collectors, Scrap Traders, and Recycling Businesses.

Vision

To become the digital operating system of the scrap industry by helping businesses manage collections, inventory, sales, cash flow, and profitability from a single platform.

Mission

Replace notebooks, memory, and scattered records with a simple digital workflow designed specifically for scrap businesses.

Tagline

Track Every Kilogram. Track Every Rupee.

2. Problem Statement

Current Industry Situation

Most scrap businesses operate using:

- Memory
- Notebooks
- WhatsApp
- Basic Excel Sheets

This creates:

- Poor inventory visibility

- Unknown profit margins
 - Cash flow confusion
 - Lost business records
 - Operational inefficiencies
-

Major Problems

Problem 1 – Profit Uncertainty

Owners don't know:

- Daily profit
 - Weekly profit
 - Monthly profit
-

Problem 2 – Inventory Confusion

Owners don't know:

- What materials are available
 - Inventory value
 - Unsold stock
-

Problem 3 – Batch Traceability

Scrap collected over multiple days gets mixed together.

Result:

- Difficult profit calculation
 - No source tracking
-

Problem 4 – Cash Management

No clear visibility of:

- Fuel expenses
 - Collection costs
 - Daily spending
 - Available working capital
-

Problem 5 – Business Dependency on Memory

Critical information remains inside the owner's head.

3. Market Opportunity

Industry

Scrap Collection & Recycling Industry

Potential Customers

Individual Collectors

- Tata Ace operators
 - Pickup truck operators
 - Independent collectors
-

Small Scrap Traders

- Local scrap yards
 - Small warehouses
-

Medium Businesses

- Recycling companies
 - Scrap processing centers
-

Market Gap

Most existing software:

- Is generic
- Is expensive
- Is not designed for scrap workflows

ScrapFlow focuses specifically on scrap operations.

4. User Personas

Persona 1 – Independent Collector

Name

Rahim

Age

42

Business

Independent collector

Vehicle

Tata Ace

Pain Points

- Profit calculation
- Inventory tracking
- Cash management

Technical Skill

Low

Persona 2 – Scrap Trader

Name

Ahmed

Age

35

Employees

5

Pain Points

- Inventory management
 - Buyer tracking
 - Reports
-

Persona 3 – Yard Owner

Name

Imran

Age

48

Employees

15

Pain Points

- Multi-worker management
 - Inventory visibility
 - Profit tracking
-

5. Product Vision

Phase 1

Scrap Management SaaS

Phase 2

AI Business Assistant

Phase 3

Marketplace

Phase 4

Complete Scrap Ecosystem Platform

Collector



ScrapFlow



Buyer



Recycler

6. Customer Journey

Current Journey

Collect Scrap



Write In Notebook



Forget Records



Sell Materials



Guess Profit

Future Journey

Collect Scrap



Record In ScrapFlow



Track Inventory



Sell Materials



View Actual Profit

7. Product Goals

Goal 1

Provide complete inventory visibility.

Goal 2

Provide complete profit visibility.

Goal 3

Track every collection and sale.

Goal 4

Reduce dependence on memory.

Goal 5

Digitize scrap businesses.

8. MVP Scope

Included

Authentication

- Register
 - Login
 - Logout
-

Collection Journal

- Daily collections
 - Notes
 - Expenses
-

Batch Management

- Create batches
 - Update batches
 - Track status
-

Material Segregation

- Copper
 - Aluminum
 - Steel
 - Iron
 - Brass
 - Plastic
-

Inventory

- Add inventory

- Update inventory
 - Track inventory
-

Buyers

- Buyer database
 - Buyer history
-

Sales

- Record sales
 - Revenue tracking
-

Expenses

- Fuel
 - Transport
 - Maintenance
 - Miscellaneous
-

Profit Dashboard

- Daily
 - Weekly
 - Monthly
-

Reports

- PDF
 - Excel
-

9. Future Scope

Version 2

Voice Notes

Speech To Text

AI Assistant

WhatsApp Integration

Version 3

OCR

Price Intelligence

Route Optimization

Marketplace

10. Product Modules

Authentication

User management and security.

Collection Journal

Daily business records.

Batch Management

Track scrap collection batches.

Material Segregation

Convert mixed scrap into categorized materials.

Inventory Management

Maintain stock levels.

Buyer Management

Store buyer information.

Sales Management

Track sales and revenue.

Expense Management

Track operational expenses.

Profit Management

Automatic profit calculations.

Reporting

Generate business reports.

11. Dashboard Specification

KPI Cards

Today's Collections

Today's Sales

Today's Expenses

Today's Profit

Current Inventory Value

Total Inventory Weight

Active Batches

Unsold Inventory

12. Business Rules

Rule 1

Inventory cannot be negative.

Rule 2

Sales quantity cannot exceed inventory.

Rule 3

Every material belongs to a batch.

Rule 4

Every sale belongs to a buyer.

Rule 5

Profit calculations include all expenses.

13. User Stories

Story 1

As a collector, I want to record collection expenses so that I can track spending.

Story 2

As a collector, I want to track inventory so that I know available stock.

Story 3

As a collector, I want to record sales so that I know my revenue.

Story 4

As a collector, I want automatic profit calculations so that I understand business performance.

Story 5

As a trader, I want reports so that I can analyze my business.

14. Database Design

Users

- id
 - name
 - email
 - password
 - role
 - created_at
-

Batches

- id
- user_id
- batch_name
- date
- collection_cost
- fuel_cost
- status

Materials

- id
- batch_id
- material_type
- weight

Inventory

- id
- material_type
- available_weight

Buyers

- id
- name
- phone
- address

Sales

- id
- buyer_id
- material_id
- quantity
- price
- total_amount
- sale_date

Expenses

- id
- batch_id
- expense_type
- amount

15. Technical Architecture

Frontend

- HTML
- CSS
- JavaScript
- Bootstrap

Backend

- FastAPI

Database

- MySQL

Authentication

- JWT

Charts

- Chart.js
-

Exporting

- ReportLab
- Pandas
- OpenPyXL

16. System Architecture

Frontend

(HTML/CSS/JS)



FastAPI Backend



MySQL Database



Storage Layer



Future AI Layer

17. Folder Structure

scrapflow-ai/

frontend/

|

├─ css/

├─ js/

├─ pages/

└─ assets/

backend/

|

├─ routes/

├─ models/

├─ services/

├─ database/

├─ auth/

└─ utils/

tests/

docs/

deployment/

18. Competitive Advantage

Generic Software

- Not designed for scrap businesses
-

ScrapFlow

- Built specifically for scrap workflows
 - Batch-based tracking
 - Inventory-focused
 - Profit-focused
 - Future AI integration
-

19. Revenue Model

Free

₹0

Features:

- Single user
 - Limited records
-

Pro

₹199/month

Features:

- Unlimited records
 - Reports
 - Analytics
-

Business

₹499/month

Features:

- Team access
 - Multi-user support
 - Advanced analytics
-

20. Risk Analysis

Risk 1

Users forget to enter data.

Solution

Voice notes in V2.

Risk 2

Users don't want to pay.

Solution

Freemium model.

Risk 3

Users resist technology.

Solution

Simple mobile-first UI.

Risk 4

Poor internet connectivity.

Solution

Future offline support.

21. KPI Definitions

Revenue

Total sales value.

Expenses

Collection + Fuel + Miscellaneous.

Profit

Revenue – Expenses.

Inventory Value

Material Weight × Selling Price.

22. Success Criteria

First 30 Days

- Father uses daily.
-

First 90 Days

- 10 active users.
-

First 6 Months

- 50 users.
-

First Year

- 100 paying users.
-

23. Development Roadmap

Month 1

Foundation

- Authentication
 - Database
 - Batch Management
 - Inventory
-

Month 2

Business Operations

- Buyers
 - Sales
 - Expenses
 - Reports
 - Profit Dashboard
-

Month 3

Product Completion

- Testing
 - Optimization
 - Deployment
 - User Feedback
-

24. Final Product Statement

ScrapFlow AI is a mobile-first SaaS platform designed specifically for scrap collectors and traders. It provides complete visibility into collections, inventory, sales, expenses, and profitability through a simple workflow tailored to real-world scrap operations. Starting as a business management platform, ScrapFlow will evolve into an AI-powered operating system

